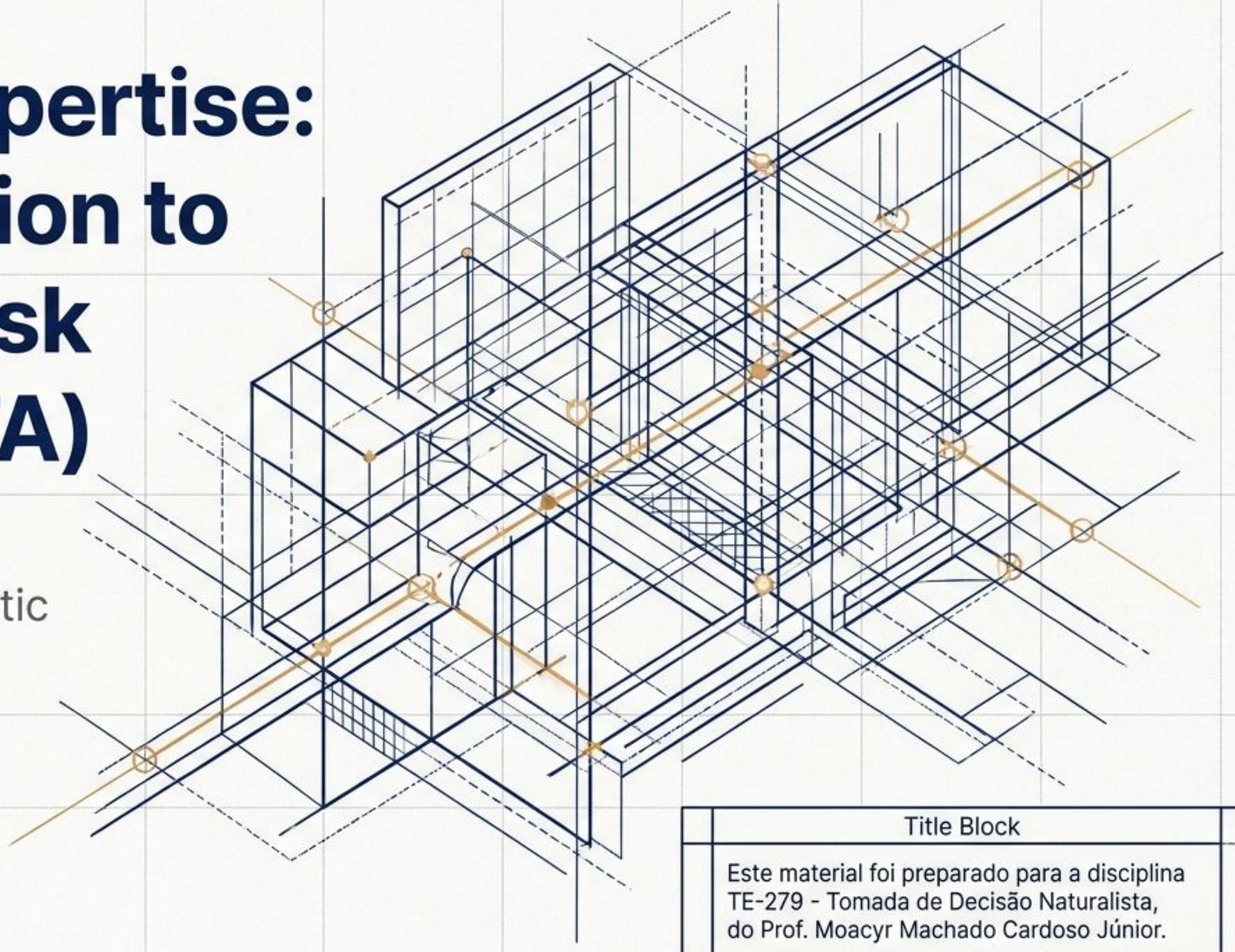
Decoding Expertise: An Introduction to Cognitive Task Analysis (CTA)

Exploring ACTA, CDM, and the Mechanics of Naturalistic Decision Making

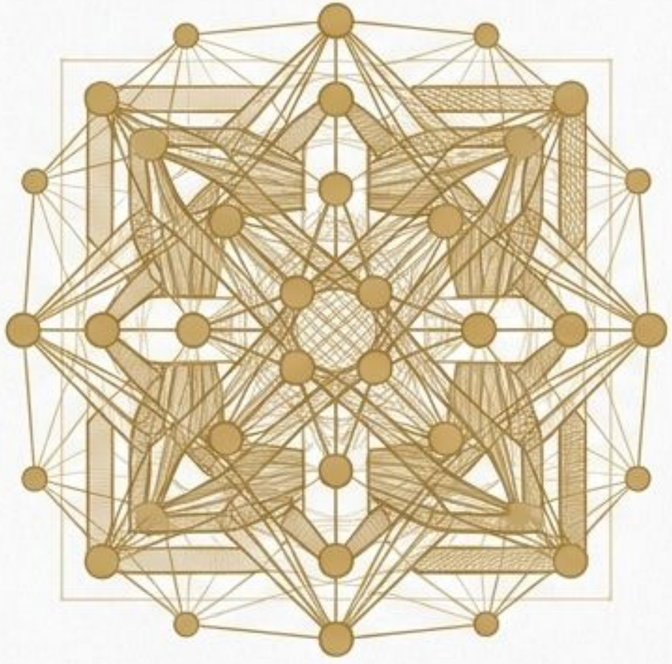


Title Block
Este material foi preparado para a disciplina TE-279 - Tomada de Decisão Naturalista, do Prof. Moacyr Machado Cardoso Júnior.

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The Paradox of Expertise

The Reality



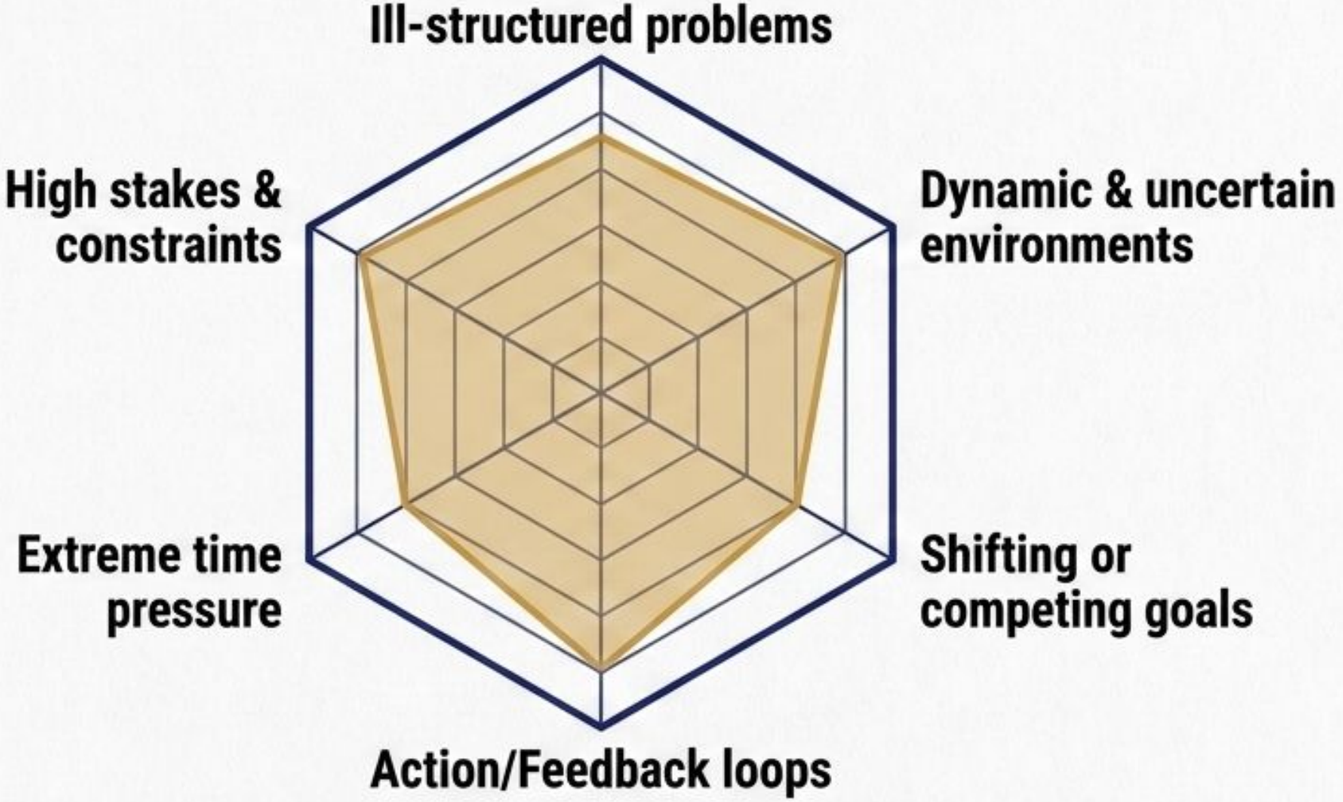
The Verbalization

As individuals gain expertise, their cognitive processes become highly automated and largely subconscious. The very experts who perform safety-critical tasks proficiently are often the least capable of verbalizing the specific rules, cues, and strategies that guide their behavior. They cannot simply tell you what they know.

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The Foundation: Naturalistic Decision Making (NDM)

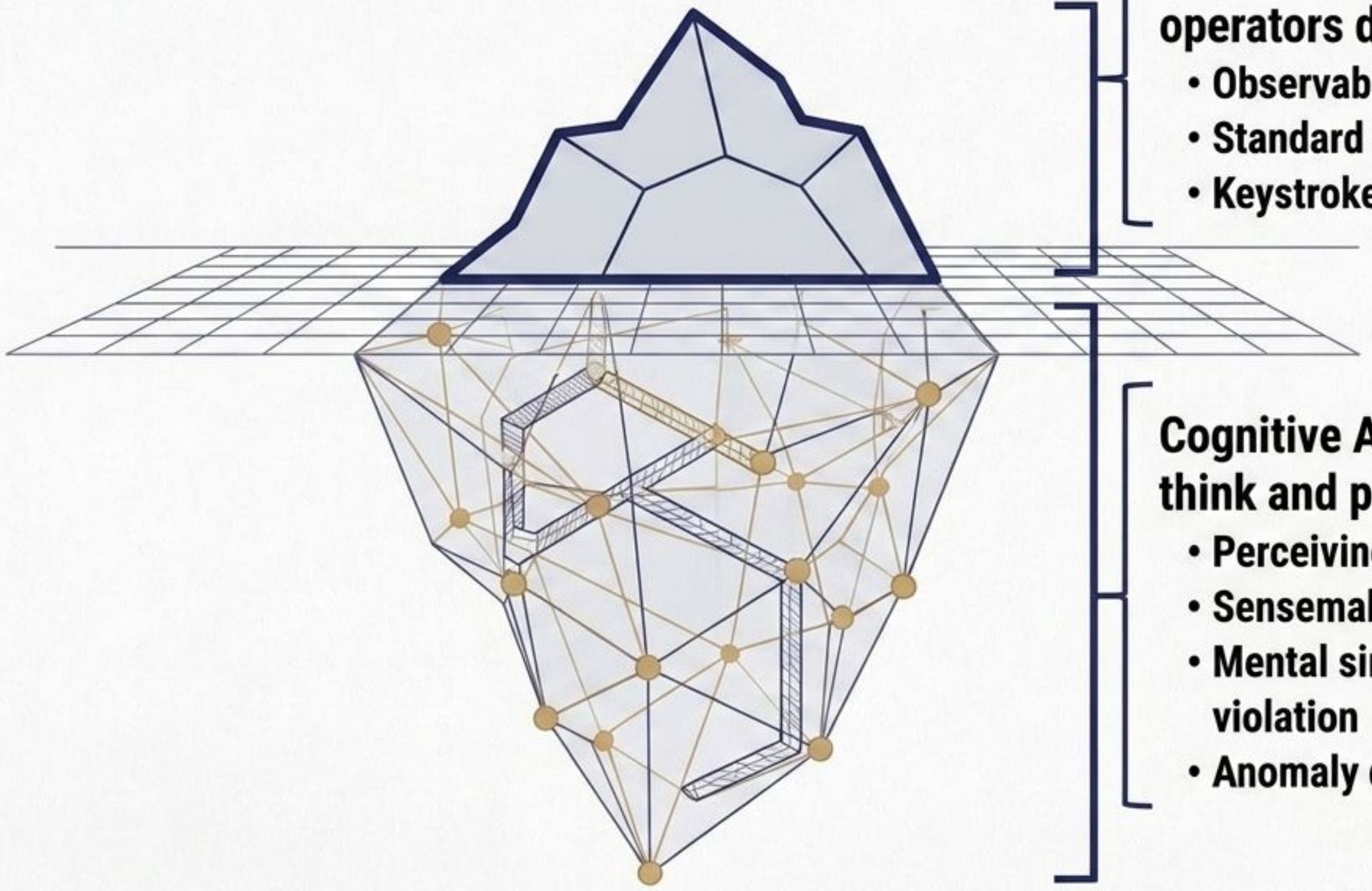
CTA is the methodological engine of NDM. Rather than studying artificial laboratory tasks, it studies how experts decide in the wild.



In these environments, experts do not compare options; they use Recognition-Primed Decision (RPD) making based on highly salient cues.

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The Iceberg: Traditional vs. Cognitive Task Cognitive Task Analysis



Traditional Analysis maps WHAT operators do:

- Observable physical actions
- Standard Operating Procedures (SOPs)
- Keystrokes and manual behaviors

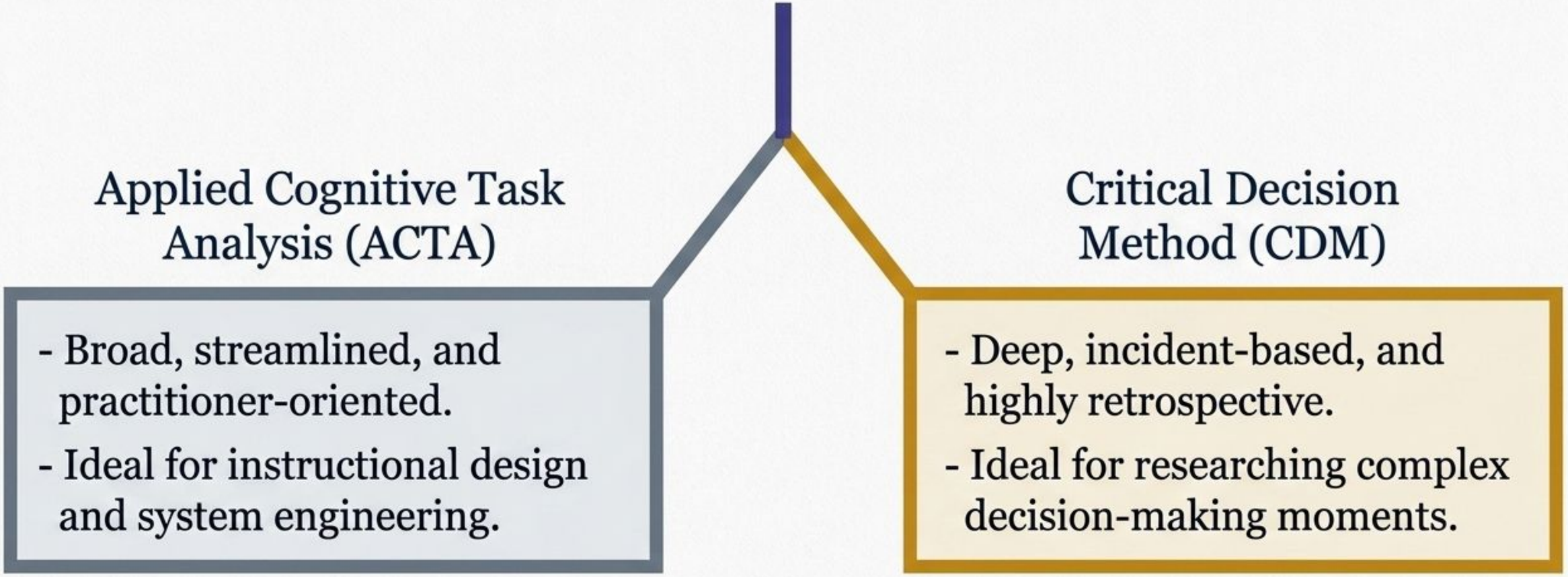
Cognitive Analysis maps HOW they think and perceive:

- Perceiving subtle cues
- Sensemaking and goal prioritization
- Mental simulation and expectancy violation
- Anomaly detection

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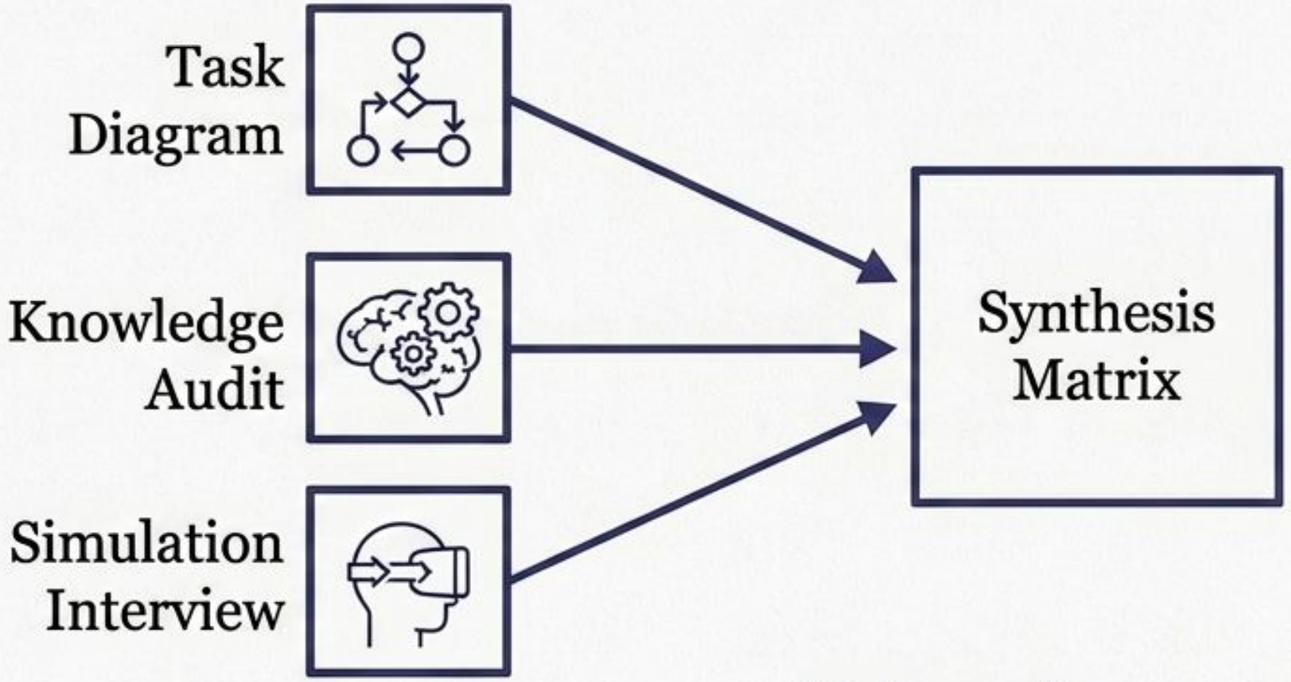
The Methodological Fork in the Road

CTA is not a single tool, but a family of elicitation techniques designed to capture tacit knowledge. For high-risk domains, two frameworks dominate the literature:



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Method 1: Applied Cognitive Task Analysis (ACTA)



History

Developed in 1998 by Militello & Hutton, ACTA was engineered to bridge the gap between academic cognitive psychology and applied system design.

Philosophy

A streamlined methodology that allows non-psychologists (instructors, UI/UX designers, engineers) to rapidly elicit and represent the critical cognitive demands of expert performance without months of data collection.

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The ACTA Pipeline (Information Architecture)

1. Task Diagram

A macro-cognitive filter. The expert decomposes the task into strictly 3-6 steps to identify “cognitive hotspots” and prevent getting bogged down in physical actions.

2. Knowledge Audit

Probing specific sub-tasks for tacit expertise. Focuses on identifying anomalies, big-picture awareness, and expert workarounds.

3. Simulation Interview

Walking the expert through a simulated scenario (e.g., using sea charts). Asking: “What actions would you take based on this cue?”

4. Synthesis

Translating the raw qualitative data into actionable system requirements and design interventions.

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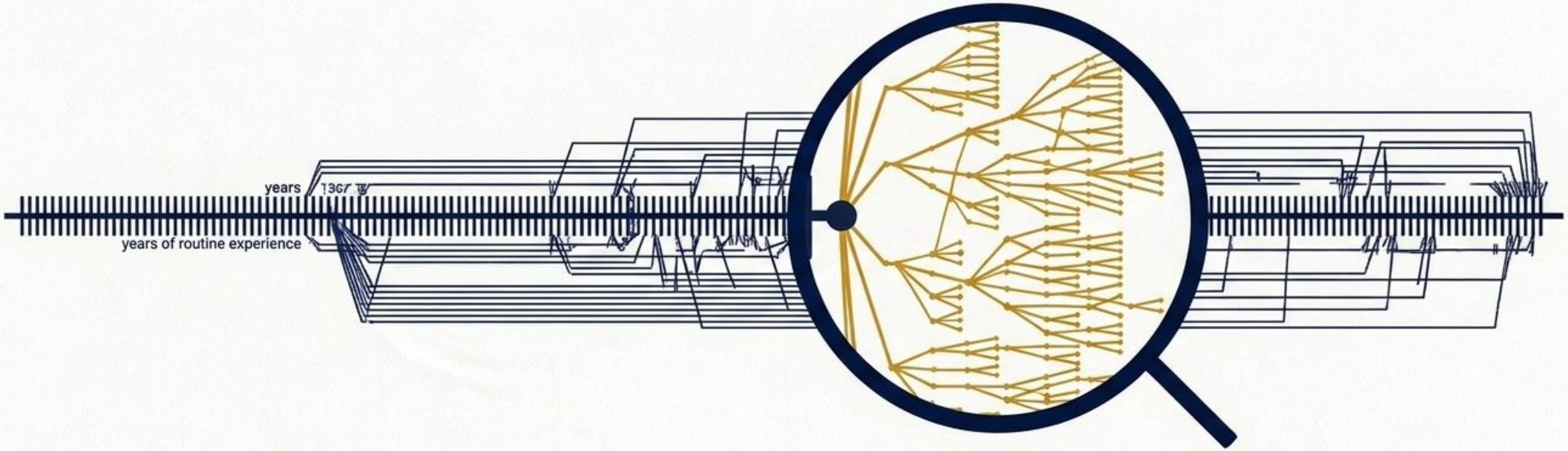
The Output: The Cognitive Demands Table (CDT)

ACTA's true power lies in its output layer. It formats complex cognitive data into structured columns for immediate applied design.

Difficult Cognitive Element	Why Difficult?	Common Errors	Cues & Strategies Used
Knowing where to search after an explosion	Novices default to starting at the source of the explosion. Not everyone knows about the Material Safety Data Sheets.	Starting at the source is a rule of thumb for most other incidents, but deadly here.	Refer to Material Safety Data Sheets for chemical locations; consider structural integrity.

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Method 2: Critical Decision Method (CDM)



CDM is a **retrospective**, semi-structured interview technique originally created to study fireground commanders. Instead of asking “how do you usually do your job?” (**which yields** generic answers), it forces the expert to **recall a specific, highly challenging, non-routine incident** where their tacit expertise prevented a failure.

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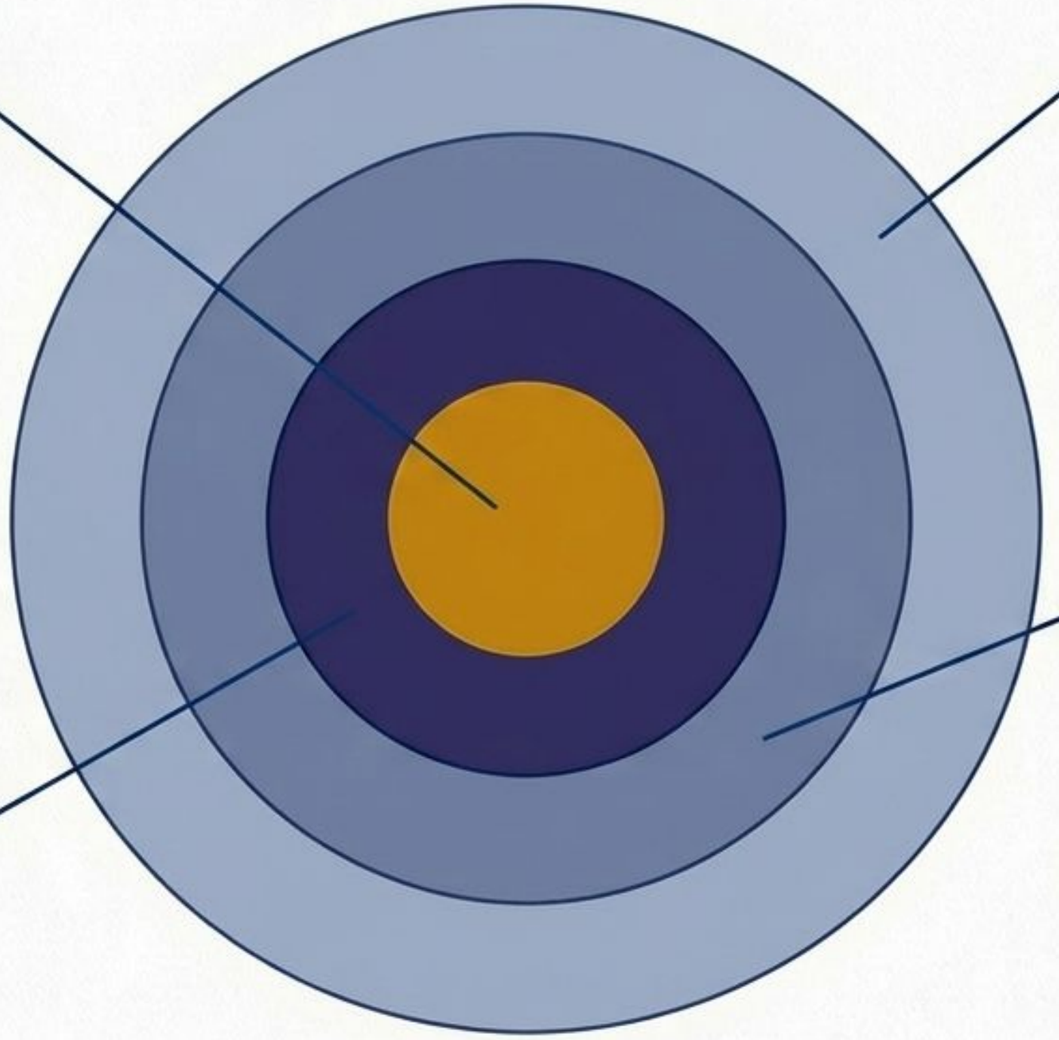
The Four Sweeps of CDM

Sweep 1: Incident Identification: The expert provides a free-form narrative of a challenging, specific event.

Sweep 2: Timeline Construction: Interviewer and expert establish a rigid chronology, identifying specific ‘Decision Points’ and shifts in situational understanding.

Sweep 3: Deepening & Probing: Applying structured cognitive probes at each Decision Point to extract the invisible cues and goals.

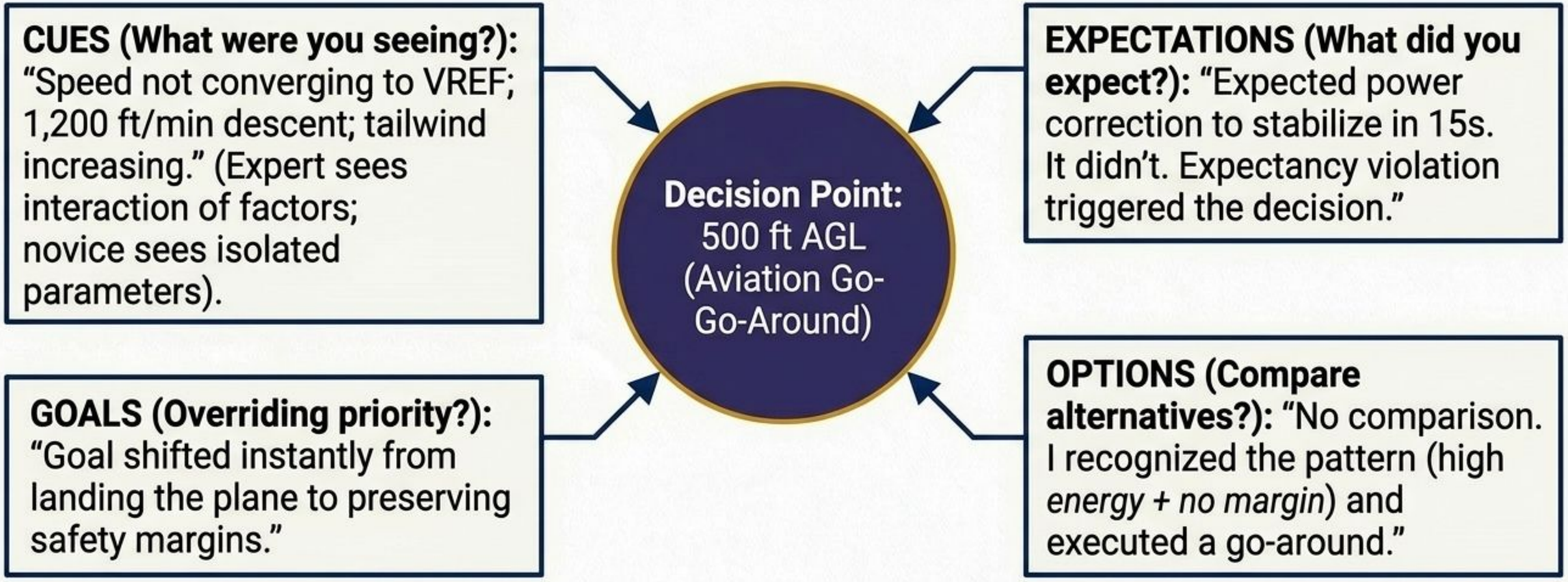
Sweep 4: What-Ifs: Exploring the boundaries of expertise through hypotheticals (‘What if this cue was missing?’).



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The Anatomy of a CDM Probe

Experts cannot answer “Why did you decide that?” CDM uses structured probes to bypass **rationalization** and uncover **Recognition-Primed Decision** elements.



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Methodological Comparison: ACTA vs. CDM

	ACTA (Applied Cognitive Task Analysis)	CDM (Critical Decision Method)
Primary Goal	Map general task domains and cognitive hotspots.	Unpack specific critical incidents and retrospective decisions.
Ideal User	Practitioners, Instructional Designers, Systems Engineers.	Researchers analyzing complex, high-stakes decision-making.
Time / Resource Burden	Moderate (Streamlined toolkit).	High (Intensive transcription and coding).
Core Output	Cognitive Demands Table (CDT).	Decision Requirements Table (DRT) & Timelines.

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Empirical Applications Across Domains



Aviation/ATC:
Informing Human Performance Modeling, interface design, and radar controller workload management.



Marine Piloting:
Utilizing ACTA and simulator interventions to map the cognitive demands of complex berthing maneuvers.



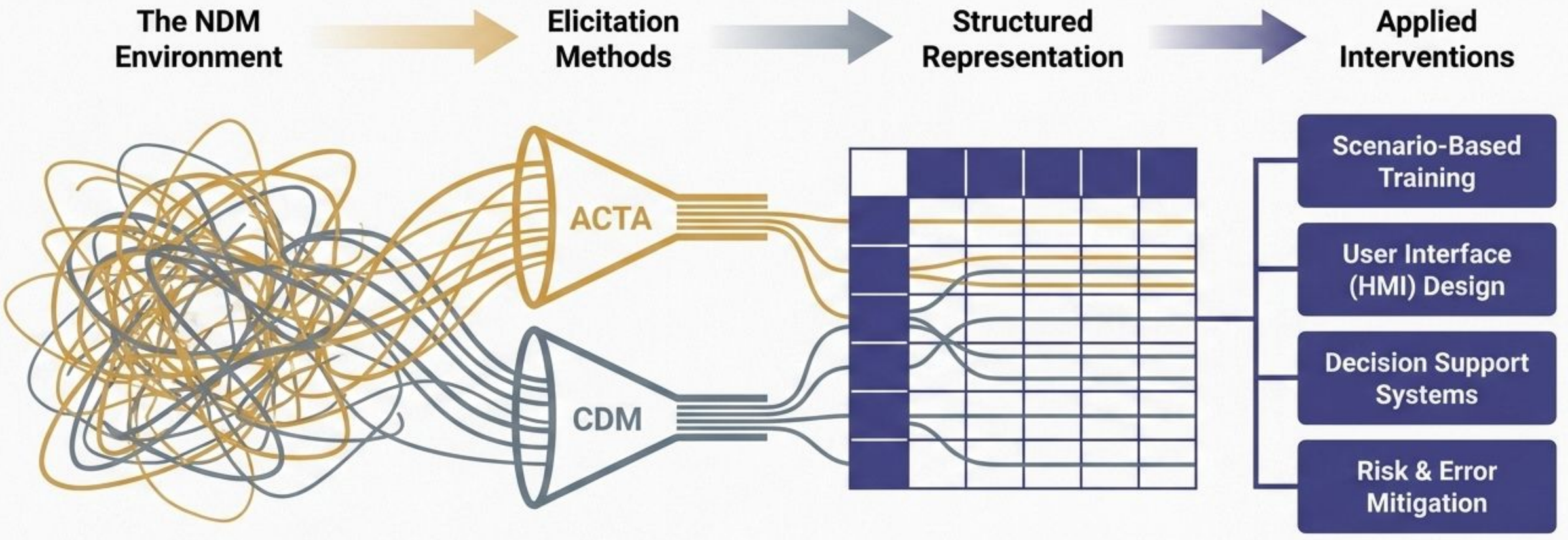
Healthcare/Nursing:
Using ACTA to capture how experienced surgical nurses detect early signs of acute patient deterioration versus novices.



Digital Forensics:
Employing CDM to map the decision-making skills of cyber-analysts tracking complex network intrusions under time pressure.

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Synthesis: The Cognitive Mapping Ecosystem



The NDM Environment

Real-world constraints: time pressure, high stakes, ambiguous cues.

Elicitation Methods

The environment is filtered. Tacit knowledge becomes explicit.

Structured Representation

Generation of Cognitive Demands Tables (CDT) and Decision Requirements Tables (DRT).

Applied Interventions

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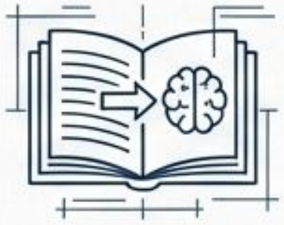
Recommended Readings



“Working Minds: A Practitioner’s Guide to Cognitive Task Analysis” (Crandall, Klein, & Hoffman, 2006). The definitive foundational text for methodology.



“Applied Cognitive Task Analysis (ACTA): A practitioner’s toolkit...” (Militello & Hutton, 1998). The primary source for the ACTA framework.



“Critical decision method for eliciting knowledge” (Klein, Calderwood, & Macgregor, 1989). The origin of CDM and its **connection** to the RPD model.



“Sources of Power: How People Make Decisions” (Klein, 1998). The seminal work on Naturalistic Decision Making.

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